

NotesNPU

Private, offline AI lecture & meeting copilot - accelerated by the Snapdragon NPU on HP PCs

Qualcomm AI Hub Whisper on the Hexagon NPU + on-device LLM | 100% offline

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<https://github.com/mdharishsuhaib/NotesNPU>

Taking notes while learning is broken

- Indian students attend 25-30 hours of lectures a week, often in fast, code-mixed Hindi-English, and struggle to take notes and listen at the same time.
- Cloud transcription tools (Otter, Fireflies, cloud Copilots) upload sensitive classroom and meeting audio to third-party servers.
- They need reliable internet and paid subscriptions, which are hard to get in hostels, tier-2/3 colleges, trains and field work.
- Running AI on the CPU drains the battery and heats the laptop, so it can't run through a whole day of classes.

25-30 h

lectures / week per student

0

bytes of audio should leave the laptop

1 day

battery needed for back-to-back classes

NotesNPU: your private AI note-taker

- Record or upload a lecture and get a timestamped transcript, fully on-device, even in airplane mode.
- Qualcomm AI Hub Whisper runs on the Hexagon NPU via ONNX Runtime QNN EP, leaving the CPU free and saving battery.
- A local LLM (Phi-3.5-mini INT4) generates a summary, structured notes, flashcards, a quiz and answers questions.
- Auto-extracts action items and deadlines ('exam next Monday', 'read chapter 6').
- Supports 10+ Indian languages, with Hindi lecture to English notes translation in the same pass.

Transcribe

Upload or record; pause-aware 30 s chunking; timestamps

Live mode

Mic streaming, text every ~8 s, on-device

Translate

Hindi / Tamil / Telugu / Bengali... to English

Study notes

Summary, key concepts, detailed notes, deadlines

Revise

Flashcards and 5-question MCQ quiz

Ask the lecture

Grounded Q&A chat on your transcript

Export

One-click Markdown notes

Benchmark

Built-in NPU vs CPU performance panel

Live demo: lecture in, study notes out

NotesNPU

Private, offline AI lecture & meeting copilot - accelerated by the Snapdragon NPU

Device: Intel(R) Core(TM) i5-7200U CPU @ 2.50GHz | **NPU not detected** | Speech: ONNX Whisper | CPU | Study AI: phi-3.5-mini-instruct-int4 | onnxruntime-genai (CPU) | 100% offline - audio never leaves this PC

1. Transcribe 2. Live mode 3. Study notes 4. Ask the lecture 5. NPU benchmark

Lecture / meeting audio

0:00 1:25

Spoken language

English

Translate to English (e.g. Hindi lecture -> English notes)

Load sample lecture

Transcribe on-device

Transcript (timestamped)

[00:00] Good morning everyone. Today we will talk about photosynthesis, the process plants use to make food. Photosynthesis happens mainly in the leaves, inside tiny structures called chloroplasts. Chloroplasts contain a green pigment called chlorophyll, which absorbs sunlight. The overall reaction is simple, carbon dioxide plus water, using light energy, produces glucose and oxygen.

[00:27] There are two main stages. The first stage is the light dependent reactions, which take place in the thylacoid membranes. Here, light energy splits water molecules, releasing oxygen and produces ATP and NADPH. The second stage is the Calvin cycle, which happens in the stroma. The Calvin cycle uses ATP and NADPH to fix carbon dioxide into glucose.

[00:55] 3. Factors affect the rate of photosynthesis, light intensity, carbon dioxide concentration, and temperature. Remember, for your exam next Monday, you must be able to draw and label a chloroplast, and explain why photosynthesis is important for life on Earth. It produces the oxygen we breathe and is the base of almost every food chain. Please read Chapter 6 before the next class. Thank you.

Performance

```

1  {
2    "backend": "ONNX Whisper | CPU",
3    "audio_s": 84.77,
4    "processing_s": 19.82,
5    "real_time_factor": 0.234,
6    "speed_x_realtime": 4.3,
7    "encoder ms avg": 2147.4.

```

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1. Transcribe 2. Live mode 3. Study notes 4. Ask the lecture 5. NPU benchmark

Summary

Structured notes

Flashcards

Quiz

Key Concepts

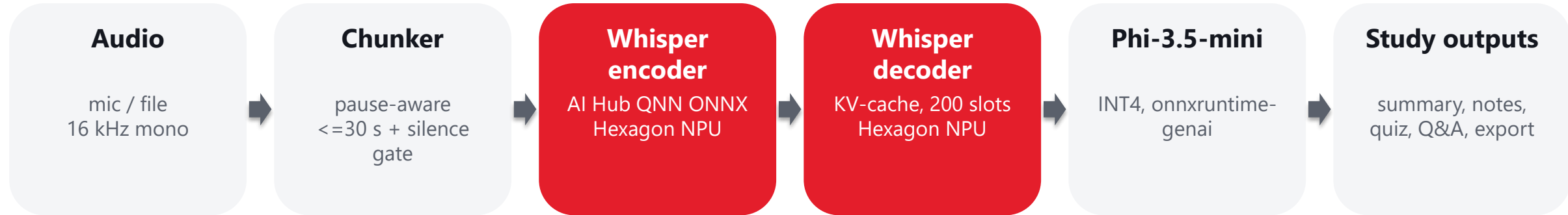
- Photosynthesis:** Process by which plants convert light energy into chemical energy (glucose) using CO2 and H2O.
- Chloroplasts:** Organelles within plant cells where photosynthesis occurs, containing chlorophyll.
- Light Dependent Reactions:** Stage involving conversion of light energy into chemical energy (ATP and NADPH).
- Calvin Cycle:** Stages that utilize ATP and NADPH to synthesize glucose from CO2.
- Factors Affecting Rate:** Light intensity, CO2 concentration, and temperature influence photosynthesis efficiency.

Detailed Notes

- Location of Photosynthesis:** Mainly in leaf chloroplasts.
 - Occurs in Thylakoid Membranes during Light Dependent Reactions.
 - Splitting Water Molecules: Oxygen released as a byproduct.
 - Produces Energy Carriers: ATP & NADPH.
 - In Stroma during Calvin Cycle.
 - Fixation of CO2: Formation of Glucose.
- Importance of Photosynthesis**
 - Provides Oxygen: Essential for respiration.
 - Foundation of Food Chains: Primary source of organic matter.
- Exam Preparation**
 - Draw and Label a Chloroplast: Understand its structure and function.
 - Explain Photosynthesis Significance: Discuss its role in ecosystems.
- Study Material**

1. Load or record a lecture > 2. Transcribe on-device (NPU) > 3. Summary / notes / flashcards / quiz > 4. Ask & export

Architecture: everything runs on the PC



- ONNX Runtime QNN Execution Provider (plugin EP, HTP backend, burst perf mode) runs AI Hub precompiled context binaries, so there is no on-device compile.
- Encoder emits cross-attention KV caches directly; the fixed-shape decoder loop feeds self-attention caches back, so every step stays on the NPU.
- Robust by design: NPU to CPU to extractive fallback chain, repetition guard, silence gating, head+tail context fitting for long lectures.

Why the Snapdragon NPU

Task	Model	Source	Runtime / hardware
Speech-to-text (NPU)	Whisper-Base, precompiled QNN ONNX FP16	Qualcomm AI Hub	ONNX Runtime + QNN EP (HTP) on the Hexagon NPU
Speech-to-text (fallback)	Whisper-Base ONNX (INT8 decoder)	Hugging Face onnx-community	ONNX Runtime CPU EP
Study assistant	Phi-3.5-mini-instruct INT4 AWQ	Microsoft (Hugging Face)	onnxruntime-genai

- Whisper encoder = dense transformer over 1500 frames, a good fit for the Hexagon HTP's FP16 tensor engines (measured: 45.5 ms per 30 s of audio on X Elite).
- AI Hub precompiled context binaries: no on-device compile, instant start-up.
- Fixed-shape KV-cache decoder (200 slots) keeps every step on the NPU (3.8 ms/token).
- NPU does the speech, so the Oryon CPU stays free for the LLM and UI and live mode never stutters.
- Low-power NPU inference means a full day of lectures on battery, quiet and cool.

Performance: measured on a real Snapdragon NPU

Device / backend	Speed	Encoder per 30 s	Decoder per token
Intel i5-7200U laptop (dev machine), CPU	5.7x real time (14.9 s)	1312 ms	44.8 ms
Snapdragon X Elite NPU (measured on Qualcomm AI Hub)	~80x real time (~1.1 s)	45.5 ms	3.8 ms
NPU speed-up	~14x end to end	28.8x	11.8x

- Measured on a real Snapdragon X Elite (Qualcomm AI Hub hosted device): encoder 45.5 ms per 30 s of audio, decoder 3.8 ms per token.
- 100% of ops on the NPU: 556 encoder ops + 975 decoder ops on the Hexagon HTP, none on the CPU; peak memory 34 MB / 60 MB.
- Accuracy: NPU (FP16) encoder output matches the FP32 CPU reference with cosine similarity 0.999, and gives an identical, correct transcript of the sample lecture.
- The 85 s sample lecture takes ~1.1 s on the NPU vs 14.9 s on a laptop CPU.

Who it helps

Students

Focus on understanding, not scribbling; revise with auto flashcards and quizzes.

Accessibility

Live captions for deaf and hard-of-hearing learners; notes for students with dyslexia or ADHD.

Regional language learners

Hindi / regional lectures turned into English notes, bridging the medium-of-instruction gap.

Professionals

Confidential meetings summarised with action items, and nothing is uploaded.

Teachers

Publish lecture notes and question banks from their own recordings.

Deployment & accessibility

- One-click setup.bat / run.bat: detects Snapdragon, installs native ARM64 deps, downloads AI Hub models.
- Runs as a local web app on 127.0.0.1, with offline mode enforced (HF_HUB_OFFLINE, no analytics).
- Automatic fallback: NPU to CPU to extractive engine, so it works on every Windows PC.
- Light install (-NoLLM, ~300 MB) for low-storage laptops.
- Open source (MIT) on GitHub, with CLI tools for transcription and benchmarking.

```
> git clone <repo>  
> cd notesnpu  
> setup.bat  
> run.bat
```

Opens <http://127.0.0.1:7860>
Works offline after setup

Roadmap

- LLM on the NPU: AI Hub Llama-3.2-3B / Phi-3.5 QNN binaries via Genie
- Speaker diarisation for meetings (who said what)
- Slide / whiteboard capture with on-device OCR, linked to the timeline
- Semantic search across a semester of lectures (on-device embeddings)
- MSIX installer on the Microsoft Store; HP AI Companion integration

Thank you

NotesNPU: learn more, write less, and keep it on your PC.

<https://github.com/mdharishsuhaib/NotesNPU>
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